

Algebra: Chapter 0, Aluffi

Part 5: Irreducibility and Factorization in Integral Domains

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§1 Chain Conditions and Existence of Factorizations

3.

(a)

The action of $k[t]$ on $k[x]$ is determined by the ring homomorphisms $k[t] \xrightarrow{\varphi} R \hookrightarrow k[x]$; consequently, the action is entirely by the action of constants in k and the indeterminate t :

$$\begin{aligned} a \cdot g(x) &= ag(x) & (a \in k) \\ t \cdot g(x) &= f(x)g(x) \end{aligned}$$

Let $f(x) = a_n x^n + \dots + a_0$. The definitions above indicate that $1, x, \dots, x^{n-1}$ generate $k[x]$ as a $k[t]$ -module: as a ring, $k[x]$ is generated by $1, x, x^2, \dots$, so it suffices to show that one can retrieve all powers of x by the action of $k[t]$ on polynomials in x of degree less than n . Indeed,

$$x^n = a_n^{-1} \cdot (t \cdot 1 - a_{n-1} \cdot x^{n-1} - \dots - a_0 \cdot 1) \quad (1)$$

shows $x^n \in \langle 1, x, \dots, x^{n-1} \rangle$, and induction and a similar formula as (1) shows all powers of x are generated by $\langle 1, x, \dots, x^{n-1} \rangle$.

(b)

$k[x]$ is a finitely generated $k[t]$ -module and $k[t]$ is a Noetherian ring (not yet covered in the book; polynomial rings over fields are PIDs), so $k[x]$ is a Noetherian $k[t]$ -module. Hence to prove R is finitely generated, it suffices to check that R is a $k[t]$ -submodule of $k[x]$. Indeed, because R contains the element f and field k , for any $p(t) \in k[t]$ and any $g[x] \in R$,

$$p(t) \cdot g = p(f)g \in R.$$

(c)

k itself is a field, so it is a Noetherian ring. Take a subring of $k[x]$ that properly contains k ; that is, consider $R \subset k[x]$ as in the last two parts, its $k[t]$ -module structure induced by some nonconstant polynomial $f \in R$. By part (b), R is a Noetherian $k[t]$ -module. Take an ideal $I \subset R$. It is clear that I is a $k[t]$ -module of R , so

I is finitely generated as a $k[t]$ -module. Suppose $I = \langle a_1, \dots, a_n \rangle$ as a $k[t]$ -module. For all $a \in I$, there exists $p_1(t), \dots, p_n(t) \in k[t]$ such that

$$\begin{aligned} a &= p_1(t) \cdot a_1 + \dots + p_n(t) \cdot a_n \\ &= p_1(f)a_1 + \dots + p_n(f)a_n \end{aligned}$$

Notice that the third equality expresses a as a R -linear combination of $a_1, \dots, a_n$ (recall the action of $k[t]$ on R ; $p_1(f), \dots, p_n(f)$ are elements of R). Consequently, $I = \langle a_1, \dots, a_n \rangle$ as an R -module; that is, $I = \langle a_1, \dots, a_n \rangle$ as an ideal in R , hence I is finitely generated. This proves R is Noetherian.

8.

Let $\mathcal{F}$ be as defined in the hint, and assuming $\mathcal{F} \neq \emptyset$, let I_0 be the maximal element of $\mathcal{F}$. Since I_0 is not prime, find $a, b \notin I_0$ such that $ab \in I_0$. Since $A := \langle I_0, a \rangle$ and $B := \langle I_0, b \rangle$ properly contain I_0 , A and B both contain a product of prime ideals; say $P_1 \cdots P_t \subset A$ and $Q_1 \cdots Q_s \subset B$ for prime ideals P_i and Q_j . Then notice

$$AB = \langle I_0, ab \rangle = \langle I_0 \rangle$$

and

$$P_1 \cdots P_t Q_1 \cdots Q_s \subset AB,$$

hence I_0 contains a product of prime ideals, contradicting the construction of $\mathcal{F}$. It follows $\mathcal{F} = \emptyset$.

10.

It is clear that R/I is Artinian due to the inclusion-preserving bijection between ideals in R containing I and ideals in R/I . If R is an Artinian integral domain, the chain

$$\langle a \rangle \supset \langle a^2 \rangle \supset \langle a^3 \rangle \supset \dots$$

for any $a \in R \setminus 0$ shows that $\langle a^n \rangle = \langle a^{n+1} \rangle$ for some n , i.e., $a^n = ua^{n+1}$ for a unit $u \in R$. The cancellation law applies because R is an integral domain, giving $1 = ua$. u is the inverse of a , thus proving that R is a field.

To show that an Artinian ring has Krull dimension 0, one must show if $\mathfrak{p}, \mathfrak{p}'$ are prime ideals in R ,

$$\mathfrak{p} \subset \mathfrak{p}' \implies \mathfrak{p} = \mathfrak{p}'.$$

Indeed, one sees this is so by mapping $\mathfrak{p}$ and $\mathfrak{p}'$ to $R/\mathfrak{p}$. Using bar notation, $\bar{\mathfrak{p}} = \bar{0}$ and $\bar{\mathfrak{p}'}$ is a prime ideal in the quotient $R/\mathfrak{p}$. But the quotient ring is a field: that $\mathfrak{p}$ is prime means $R/\mathfrak{p}$ is an integral domain, and $R/\mathfrak{p}$ Artinian. Fields have no nontrivial, nonproper ideals, and since $\bar{\mathfrak{p}} \neq R/\mathfrak{p}$, it follows $\bar{\mathfrak{p}'} = \bar{0}$. This proves $\mathfrak{p}' = \mathfrak{p}$, as desired.

§2 UFDs, PIDs, Euclidean Domains

6.

For any $a, b \in R \setminus 0$, there exists $c \in R$ such that $\langle c \rangle = \langle a \rangle \cap \langle b \rangle$. By definition of the least common multiple, c is the LCM of a and b : if for any $m \in R$, $\langle m \rangle \subset \langle a \rangle$ and $\langle m \rangle \subset \langle b \rangle$, then $\langle m \rangle \subset \langle c \rangle$.

ab/c is then the GCD of a and b . Suppose $\langle a \rangle \subset \langle e \rangle$ and $\langle b \rangle \subset \langle e \rangle$. ab/e is then a common multiple of a and b , so $\langle ab/e \rangle \subset \langle c \rangle$. In other words, for some $q \in R$,

$$cq = ab/e \implies ceq = ab \implies eq = ab/c \implies e \mid ab/c,$$

verifying ab/c is the gcd of a and b .

7.

For any elements $a, b \in R \setminus 0$, let $d = ra + sb$ be the GCD of a and b . Namely, the fact that d is a R -linear combination means

$$\langle a, b \rangle = \langle d \rangle. \quad (2)$$

R being Noetherian means every ideal $I \triangleleft R$ is finitely generated. But by (2), one can induct on the number of generators to show that any finitely generated ideal is in fact generated by a single element; this shows R is a PID.

9.

Let $\mathfrak{p}$ be a prime ideal in a UFD R of height 1. Namely, since 0 is a prime ideal in any integral domain, the height of $\mathfrak{p}$ says $\mathfrak{p}$ does not properly contain any nonzero prime ideal of R . Take any $a \in \mathfrak{p} \setminus 0$. Let $a = p_1 \cdots p_k$ be the prime decomposition for a . Since $\mathfrak{p}$ is prime, $a \in \mathfrak{p}$ means some $p_i \in \mathfrak{p}$, i.e.,

$$\langle p_i \rangle \subset \mathfrak{p}.$$

By the height of $\mathfrak{p}$, this inclusion must, in fact, be an equality. Hence $\mathfrak{p}$ is principal.

12.

Let $R[x]$ be a PID and take any $a \in R \setminus 0$. By assumption, there exists $f(x) \in R[x]$ such that

$$\langle a, x \rangle = \langle f \rangle. \quad (3)$$

(3) says $f \mid x$, but because $R[x]$ is an integral domain, necessarily $f = 1$ or $f = x$ (up to units). Furthermore, $f \mid a$ means $f \neq x$ since, once again, $R[x]$ is an integral domain. So $f = 1$, meaning $\langle a, x \rangle = \langle 1 \rangle$. Accordingly, there are $g(x), h(x) \in R[x]$ such that

$$ag + xh = 1$$

Namely, xh is a polynomial with no constant term, so, letting c be the constant term of g ,

$$ac = 1.$$

Namely, this shows that a is a unit in R . This proves R is a field.

13.

The proof of $x^a - 1 \mid x^b - 1 \iff a \mid b$ follows the same way as the proof of $c^a - 1 \mid c^b - 1 \iff a \mid b$; I will only show the latter. If $a \mid b$, then $b = aq$ for some q .

$$c^b - 1 = c^{aq} - 1 = (c^a - 1)(c^{a(q-1)} + c^{a(q-2)} + \dots + c^a + 1)$$

shows $c^a - 1 \mid c^b - 1$.

Conversely, suppose $c^a - 1 \mid c^b - 1$. Write $b = aq + r$ for some $0 \leq r < a$. $c^a - 1 \mid c^{aq} - 1$ always, so

$$c^a - 1 \mid [c^{aq}c^r - 1 - (c^{aq} - 1)] = c^{aq}(c^r - 1)$$

Since $c^a - 1 \mid c^{aq-1}$, $\gcd(c^a - 1, c^{aq}) = 1$, which forces

$$c^a - 1 \mid c^r - 1. \quad (4)$$

But since $r < a$, (4) is nonsense unless $c^r - 1 = 0$, i.e., unless $r = 0$. This proves $a \mid b$.

17.

Take any nonunit $c_1 \in R \setminus 0$. If c_1 is not the desired element, there is an element $a_1 \in R$ such that $a_1 = c_1q_1 + r_1$ for some nonunit r_1 such that $r_1 \neq 0$, $v(r_1) < v(c_1)$. Let $c_2 = r_1$, and once again, if c_2 is not the desired element, find $a_2 \in R$ such that $a_2 = c_2q_2 + r_2$ for some nonunit, nonzero r_2 with $v(r_2) < v(c_2)$, and repeat the process for $c_3 = r_2$, etc. This process must eventually terminate, yielding some c_k such that for all $a \in R$, there is some $r, q \in R$ such that $a = c_kq + r$ and r is zero or a unit. If not, since $v(r_i) < v(c_i)$ for each c_i that doesn't satisfy the desired property, the v will eventually start mapping to the negative numbers, which contradicts the fact that the codomain of v is the nonnegative integers.

§3 Intermezzo: Zorn's Lemma**11.**

Suppose every nonzero prime ideal is a maximal ideal. Then every maximal ideal is a principal ideal, which in turn shows that every prime ideal is a principal ideal: let $\mathfrak{m}$ be a maximal ideal and pick any $a \in \mathfrak{m}$. Let $a = q_1 \cdots q_k$ be the irreducible decomposition of a . Since $\mathfrak{m}$ is a maximal ideal (namely, it is prime), some $q_i \in \mathfrak{m}$. But irreducible elements are prime elements in a UFD and prime elements generate a maximal ideal by the hypothesis on R , hence $\langle q_i \rangle = \mathfrak{m}$. This shows that maximal ideal are principal.

Let

$$\mathcal{F} = \{I \neq 0 \not\subseteq R \mid I \text{ is not a principal ideal}\}.$$

One can easily check that every chain in $\mathcal{F}$ has an upper bound in $\mathcal{F}$; by Zorn's lemma, $\mathcal{F}$ has a maximal element I_0 . Since I_0 is not principal, it is necessarily not prime. Take $a, b \notin I_0$ such that $ab \in I_0$. $I_0 \subsetneq \langle I_0, a \rangle$ and the maximality of I_0 in $\mathcal{F}$ means $\langle I_0, a \rangle = \langle a' \rangle$ for some $a' \in R$. Similarly, $\langle I_0, b \rangle = \langle b' \rangle$ for some $b' \in R$. But then

$$\langle a'b' \rangle = \langle a' \rangle \langle b' \rangle = \langle I_0, ab \rangle = \langle I_0 \rangle,$$

which contradicts the assumption that $I_0 \in \mathcal{F}$.

§4 Unique Factorization in Polynomial Rings

4.

The first statement is a tautology since every prime ideal in a PID is a prime principal ideal. However, primitive doesn't imply very primitive in a general UFD. Consider $R = \mathbb{Z}[x_1]$ and $f = 2x_1x_2 + 5 \in R[x_2]$. Then $\gcd(2x_1, 5) = 1$ shows f is primitive, but $2 \notin \langle 2x_1, 5 \rangle$ shows f is not very primitive.

8.

Clearly $\sim$ is symmetric and reflexive. To check that it is transitive, suppose

$$\begin{aligned} t_1(s'm - sm') \\ t_2(s''m' - s'm'') \end{aligned}$$

Then

$$\begin{aligned} t_1t_2s'(s''m - sm'') &= t_1t_2(s''s'm - s''sm' + s''sm' - s'sm'') \\ &= 0 \end{aligned}$$

shows $(m, s) \sim (m', s')$ and $(m', s') \sim (m'', s'')$ imply $(m, s) \sim (m'', s'')$.

The natural action of $S^{-1}R$ on $S^{-1}M$ that makes the latter a $S^{-1}R$ -module is

$$r/s \cdot m/t := (rm)/(st),$$

where rm is, of course, induced by the R -module structure of M . It is easy to check that this operation satisfies the axioms of a $S^{-1}R$ module, should the operation be well defined. Indeed, it is: for example, suppose $r_1/s_1 = r_2/s_2$ in $S^{-1}R$ with $u(s_2r_1 - s_1r_2) = 0$ for some $u \in S$. It must be shown that $r_1/s_1 \cdot m/t = r_2/s_2 \cdot m/t$ for any $m/t \in S^{-1}M$. This is verified by the computation

$$\begin{aligned} u(r_1s_2t \cdot m - r_2s_1t \cdot m) &= tu(s_2r_1 - s_1r_2) \cdot m \\ &= 0 \cdot m = 0. \end{aligned}$$

9.

(a)

Take $a_1/s_1, a_2/s_2 \in I^e$ and any $r/t \in S^{-1}R$. Then

$$\frac{a_1}{s_1} + \frac{r}{t} \frac{a_2}{s_2} = \frac{ta_1s_2 + ra_2s_1}{ts_1s_2} \quad (5)$$

Since I is an ideal, $ta_1s_2 + ra_2s_1 \in I$, so (5) is in I^e . This shows I^e is an ideal in $S^{-1}R$.

To verify that I^e is not proper, suppose $1 \in I^e$, i.e., there exists some $a/s \in I^e$ and some $u \in S$ such that

$$u(a - s) = 0.$$

In other words, $au = us$. But $au \in I$ while $us \in S$, contradicting the assumption $I \cap S = \emptyset$.

(b)

Take $a, b \in J^c$ and $r \in R$. Since J is an ideal, $(a + rb)/1$ in $S^{-1}R$ is an element of J , hence $a + rb \in J^c$. Further, since J is a proper ideal, $S \cap J^c = \emptyset$; for if there was some $s \in S \cap J^c$, then J contains the unit s in $S^{-1}R$, hence $J = \langle 1 \rangle$.

(c)

$(J^c)^e = J$ is clear; see Q14 for an argument similar to proving $(J^c)^e = J$.

Take $a \in (I^e)^c$. Then $a/1 \in I^e$, so $\exists a'/s \in I^e$ and $u \in S$ such that

$$u(sa - a') = 0$$

which means $asu = a'u$, with the right hand side in I . Of course, $su \in S$, so a satisfies the desired property.

Vice versa, suppose $a \in R$ such that $sa \in I$ for some $s \in S$. Then $as/s = a \in I^e$, hence $a \in (I^e)^c$.

This gives the relation $I \subset (I^e)^c$.

(d)

As given in the hint, let $R = \mathbb{C}[x, y]$ and $S = \{1, x, x^2, \dots\} \subset R$. If $I = \langle xy \rangle$, then by (c), $y \in (I^e)^c$, but $y \notin I$ clearly. This phenomenon occurs because I^e contains elements of the form

$$xyf(x, y)/x^k, \quad (f(x, y) \in R, k \in \mathbb{Z}_{\geq 0}),$$

and hence for $k = 1$, the x 's in the denominator and numerator cancel out.

10.

It is clear that if $\mathfrak{q} \triangleleft S^{-1}R$ is prime, then its contraction $\mathfrak{q}^c \triangleleft R$ is prime. Suppose $\mathfrak{p} \triangleleft R$ is prime disjoint from S and take $\frac{a}{s} \in \mathfrak{p}^e$ where $b/t \notin \mathfrak{p}^e$. Then there exists $u/v \in \mathfrak{p}^e$ with $u \in \mathfrak{p}$ and $v \in S$ such that $u/v = (ab)/(st)$, i.e., there exists $k \in S$ such that

$$kstu = kvab$$

in R . The left hand side is an element of $\mathfrak{p}$ since $u \in \mathfrak{p}$. On the right hand side, $kv \in S$ means $kv \notin \mathfrak{p}$, and further $b \notin \mathfrak{p}$ since $b/t \notin \mathfrak{p}^e$. Hence $kvb \notin \mathfrak{p}$, which forces $a \in \mathfrak{p}$ by primality of $\mathfrak{p}$. This means $a/s \in \mathfrak{p}^e$, proving the primality of $\mathfrak{p}^e$.

It is clear that the assignment $\mathfrak{p} \mapsto \mathfrak{p}^e$ preserves inclusions. The paragraph above verifies that this assignment indeed maps prime ideals of R disjoint from S to prime ideals of $S^{-1}R$, and its inverse map $\mathfrak{q} \mapsto \mathfrak{q}^c$ maps prime ideals of $S^{-1}R$ to prime ideals of R disjoint from S . To see that this mapping is a bijection, notice that if $\mathfrak{p} \triangleleft R$ is a prime ideal disjoint from S , then $(\mathfrak{p}^e)^c = \mathfrak{p}$: as noted in Q9(c), $\mathfrak{p} \subset (\mathfrak{p}^e)^c$. To go the other direction, take any $r \in (\mathfrak{p}^e)^c$, i.e., $\exists s \in S$ such that $rs \in \mathfrak{p}$. Since $\mathfrak{p}$ is disjoint from S and $\mathfrak{p}$ is prime, $r \in \mathfrak{p}$.

11.

Let $\mathfrak{p} \triangleleft R$ be a prime ideal and define $S := R \setminus \mathfrak{p}$. Then S is a multiplicative set: $1 \notin \mathfrak{p}$, and $s_1, s_2 \notin \mathfrak{p} \implies s_1 s_2 \notin \mathfrak{p}$; in other words, $1 \in S$ and $s_1, s_2 \in S \implies s_1 s_2 \in S$.

The inclusion-preserving bijection between prime ideals of R contained in $\mathfrak{p}$ and prime ideals of $R_{\mathfrak{p}}$ is just the bijection established in Q10; the “prime ideals of R contained in $\mathfrak{p}$ ” detail is due to the definition $S = R \setminus \mathfrak{p}$; hence any ideal contained in $\mathfrak{p}$ is disjoint from S .

Of course, $\mathfrak{p}$ is the unique maximal element amongst prime ideals in R contained in $\mathfrak{p}$. Since the bijection established above is inclusion-preserving, $\mathfrak{p}^e \triangleleft R_{\mathfrak{p}}$ is the unique maximal ideal of R (recall maximal ideals are prime ideals).

12.

Numbering the statements as

1. $M = 0$
2. $M_{\mathfrak{p}} = 0$ for every prime ideal $\mathfrak{p} \triangleleft R$.
3. $M_{\mathfrak{m}} = 0$ for every maximal ideal $\mathfrak{m} \triangleleft R$,

it is clear $(1) \implies (2) \implies (3)$ since maximal ideals are prime ideals. To prove the contrapositive of $(3) \implies (1)$, suppose M is not the zero R -module and take any element $m \in M \setminus 0$. Let $I = \{a \in R \mid am = 0\}$. It is clear that I is an ideal in R . Let $\mathfrak{m}$ be a maximal ideal in R containing I . Then $M_{\mathfrak{m}}$ is not the zero module: on the contrary, if $M_{\mathfrak{m}}$ is the zero module, then m/s would be 0 for all $s \in R \setminus \mathfrak{m}$. In other words,

$$\exists t \in R \setminus \mathfrak{m} \text{ such that } t(m - 0) = 0,$$

which contradicts the assumption that $t \notin I$ and I is precisely the ideal containing all the elements of R that annihilate m .

14.

Note that these extensions and contractions are with respect to $M \rightarrow S^{-1}M$, where M is viewed as a R -module and $S^{-1}M$ is viewed as a $S^{-1}R$ module. Given a submodule $N \triangleleft M$, one can check

$$N^e := \{n/s \mid n \in N, s \in S\}$$

is a submodule of $S^{-1}M$. Similarly, given a submodule $\hat{N} \triangleleft S^{-1}M$,

$$\hat{N}^c := l^{-1}(\hat{N})$$

is a submodule of M l is the map $l : M \rightarrow S^{-1}M$; $m \mapsto m/1$.

Let's verify $\hat{N} = (\hat{N}^c)^e$. Take any $n/s \in \hat{N}$. Then $\frac{s}{1} \cdot \frac{n}{s} = n/1 \in \hat{N}$, so $n \in \hat{N}^c$. It immediately follows $n/s \in (\hat{N}^c)^e$. Next, take any $n/s \in (\hat{N}^c)^e$ where $n \in \hat{N}^c$ by definition. Then, one again by definition, $l(n) = n/1 \in \hat{N}$, so $\frac{1}{s} \cdot \frac{n}{1} = n/s \in \hat{N}$.

The result $\hat{N} = (\hat{N}^c)^e$ indicates every localization of a Noetherian module is Noetherian. Assume M is a Noetherian R -module and take a submodule $\hat{N} \triangleleft S^{-1}M$. $\hat{N}^c$ is a submodule of M , so it is finitely generated. This means all possible numerators of $(\hat{N}^c)^e = \hat{N}$ are generated by finitely many elements; the action of $S^{-1}R$ on $S^{-1}M$ gives all the possible denominators of $\hat{N}$, so $\hat{N}$ is finitely generated.

15.**(a)**

Let q be irreducible in R and let $q/1 = \frac{r_1 r_2}{s_1 s_2}$. Since R is a UFD and particularly an integral domain, this equality in $S^{-1}R$ is equivalent to the equality

$$s_1 s_2 q = r_1 r_2$$

in R . $q \mid r_1 r_2$ and irreducible elements are prime in UFDs, so $q \mid r_1$ or $q \mid r_2$. Without loss of generality, let $q \mid r_1$ and write $r_1 = qr'_1$. Hence

$$\begin{aligned} q/1 &= \frac{qr'_1 r_2}{s_1 s_2} \\ \implies 1 &= \frac{r'_1 r_2}{s_1 s_2} \end{aligned}$$

where the $q/1$ can be canceled because R is an integral domain $\implies S^{-1}R$ is an integral domain. This shows that r_2/s_2 is a unit, from which follows $q/1$ is a unit or an irreducible element. It is indeed possible that $q/1$ is a unit in $S^{-1}R$; for example, if $q \in S$.

(b)

Suppose a/s is irreducible in $S^{-1}R$. Since $1/s$ is a unit in $S^{-1}R$, $a/1$ is irreducible in $S^{-1}R$, and one may as well show that $a/1$ is associate to $q/1$ for some $q \in R$ irreducible. If a is irreducible in R , there is nothing more to be shown. If a is not irreducible in R , let

$$a = p_1 p_2 \cdots p_k$$

be the irreducible factorization of a . This relation holds in $S^{-1}R$ as well. However, since a is irreducible in $S^{-1}R$, either p_1 or $p_2 \cdots p_k$ is a unit in $S^{-1}R$. If $p_2 \cdots p_k$ is a unit, then a is associate to $p_1/1$ with p_1 irreducible in R . If p_1 is a unit, then $p_2 \cdots p_k$ is irreducible in $S^{-1}R$ and thus either p_2 or $p_3 \cdots p_k$ is a unit in $S^{-1}R$, and one can proceed as before to find that a is associate to some $p_i/1$.

(c)

Take any ascending chain

$$\langle r_1/s_1 \rangle \subset \langle r_2/s_2 \rangle \subset \cdots$$

in $S^{-1}R$. Since the $1/s_i$'s are all units, the chain is above is equal to $\langle r_1/1 \rangle \subset \langle r_2/1 \rangle \subset \cdots$. In R , the chain $\langle r_1 \rangle \subset \langle r_2 \rangle \subset \cdots$ stabilizes, say at index k (one must not forget that now the $\langle r_i \rangle$'s are ideals in R). So consider $\langle r_k/1 \rangle$ and $\langle r_{k+j}/1 \rangle$ in $S^{-1}R$. Take any $\frac{a}{t}r_{k+j} \in \langle r_{k+j}/1 \rangle$. Since the ideals $\langle r_k \rangle = \langle r_{k+j} \rangle$ in R , there is some $b \in R$ such that $br_k = ar_{k+j}$. Hence

$$\frac{a}{t}r_{k+j} = \frac{b}{t}r_k \in \langle r_k/1 \rangle,$$

i.e., $\langle r_k/1 \rangle = \langle r_{k+j}/1 \rangle$. This verifies that the ascending chain condition for principal ideals is satisfied in $S^{-1}R$.

Next, suppose that a/s is irreducible in $S^{-1}R$ and that $a/s \mid \frac{r_1}{t_1} \frac{r_2}{t_2}$. By part (b), a/s is associate to some $q/1$ for q irreducible in R (of course, $q/1$ is irreducible in $S^{-1}R$). To check that a/s is prime, it suffices to check that $q/1$ divides $r_1/1$ or $r_2/1$. But indeed, q does divide r_1 or r_2 in R because R is a UFD and irreducibles are prime in a UFD; the result therefore holds in $S^{-1}R$ as well. This proves that a/s is prime in $S^{-1}R$.

Since the ascending chain condition and irreducible = prime condition hold in $S^{-1}R$, it is a UFD.

16.

Q15 already shows $S^{-1}R$ is a UFD if R is a UFD. Vice versa, suppose R is Noetherian and $S^{-1}R$ is a UFD. Recall that localizations of Noetherian rings are themselves Noetherian. Take a prime ideal $\mathfrak{p} \triangleleft R$ of height 1. Suppose $\mathfrak{p} \cap S \neq \emptyset$. Then $s^k \in \mathfrak{p}$ for some power k . Since $\mathfrak{p}$ is prime, $s \in \mathfrak{p}$ and therefore $\mathfrak{p}$ contains a nonzero prime ideal $\langle s \rangle$ (since s is a prime element). $\mathfrak{p}$ is of height 1, so $\mathfrak{p} = \langle s \rangle$.

Next, assume $\mathfrak{p} \cap S = \emptyset$. $\mathfrak{p}$ corresponds to the prime ideal $\mathfrak{p}^e \triangleleft S^{-1}R$. By the inclusion-preserving bijection between prime ideals of $S^{-1}R$ and prime ideals of R disjoint from S , $\mathfrak{p}^e$ is of height 1. $S^{-1}R$ is a Noetherian UFD, so $\mathfrak{p}^e = \frac{a}{s} S^{-1}R = a S^{-1}R$ for some $a \in R$. The identity $\mathfrak{p} = (\mathfrak{p}^e)^c$ says $aR \subset \mathfrak{p}$. Furthermore, because $a/1$ is a prime element in $S^{-1}R$, a is a prime element in R : suppose $bc \in aR$. Then $\frac{b}{1} \frac{c}{1} \in a S^{-1}R$, and because $a/1$ is prime, either $b/1 \in a S^{-1}R$ or $c/1 \in a S^{-1}R$. It follows that $b \in aR$ or $c \in aR$, proving a is prime in R . **NOT NECESSARILY TRUE!!!** Hence $\mathfrak{p}$ contains a prime ideal $\langle a \rangle$, and the height of $\mathfrak{p}$ forces $\mathfrak{p} = \langle a \rangle$.

In all, any prime ideal in R of height 1 is principal. This proves R is a UFD.

20.

Let $f = a_d x^d + \dots a_0$. First, assume f is a unit with $fg = 1$, $g = b_t x^t + \dots + b_0$. Writing $fg = \sum_{k=0}^{d+t} c_k x^k$, recall

$$c_k = \sum_{i+j=k} a_i b_j \quad (6)$$

(6) indicates $a_0 b_0 = 1$, so a_0 is a unit (and so is b_0), as desired. The following shows that a_d is nilpotent. First, by the following induction,

$$a_d^{k+1} b_{t-k} = 0$$

for all $k \geq 0$. It is trivially true for $k = 0$ since $0 = c_{d+t} = a_d b_t$. Suppose the result is true up to and including $k - 1$. By (6),

$$0 = c_{d+t-k} = a_d b_{t-k} + a_{d-1} b_{t-k+1} + \dots + a_{d-k} b_t$$

Consider multiplying a_d^k to the equation. The inductive hypothesis says that multiplying by a_d^k annihilates all the summands on the right hand side except for the first summand $a_d b_{t-k}$, and of course the left hand side remains 0. Hence $a_d b_{t-k} a_d^k = a_d^{k+1} b_{t-k} = 0$, completing the induction. Namely, this result shows

$$a_d^{t+1} b_0 = 0.$$

Since b_0 is a unit, it is not a zerodivisor, which forces $a_d^{t+1} = 0$. To see that $a_1, \dots, a_{d-1}$ are nilpotent as well, notice the following transformation. a_d is nilpotent, so $1 - a_d^t$ is a unit; because f is a unit as well,

$$\begin{aligned} f(1 - a_d^t) &= f - a_d^t(a_{d-1}x^{d-1} + \dots + a_0) \\ &= a_d x^d + (1 - a_d^t)(a_{d-1}x^{d-1} + \dots + a_0) \end{aligned}$$

is a unit. (unit) – (nilpotent) is once again a unit, so one concludes from the equality of the first and last expressions above that

$$a_{d-1}x^{d-1} + \dots + a_0$$

is a unit. Now, inductively applying the same technique as above yields $a_1, \dots, a_{d-1}$ as nilpotents. This finishes the backward direction.

The forward direction is much easier. If $f = a_dx^d + \dots + a_1x + a_0$ with a_0 a unit in R and $a_1, \dots, a_d$ nilpotents in R , then f is a sum of a unit with a nilpotent (recall that the nilradical is an ideal). The sum of a unit and a nilpotent is a unit, so f is a unit.

§5 Irreducibility of Polynomials

3.

Let $f(x)$ be a polynomial with even-power terms only and suppose $h(x) \mid f(x)$. The map

$$\Phi : R[x] \rightarrow R[x]; \quad \Phi|_R = \text{id}, x \mapsto -x$$

is an automorphism $R[x] \xrightarrow{\sim} R[x]$. Since f only has even-power terms, $\Phi(f(x)) = f(x)$. Hence, writing $f(x) = h(x)q(x)$,

$$f(x) = h(-x)q(-x),$$

which shows $h(-x)$ is a factor of $f(x)$ as well.

7.

Let g be another degree d polynomial whose values agree as that of f at $d + 1$ distinct points of R . Let $h := f - g$. Then h has at most degree d but is 0 at at least $d + 1$ distinct points of R ; because R is an integral domain, h must be the zero polynomial. In other words, $f = g$; f is uniquely determined by its value of any $d + 1$ distinct points of R .

8.

As given in the hint, first assume $b_1 = b_2 = \dots = b_d = 0$ and only $b_0 \neq 0$. Define

$$g_0(x) := (x - a_1)(x - a_2) \cdots (x - a_d)$$

and let $c_0 := g_0(a_0)$. $g_0(x)$ is a degree d polynomial whose d roots are, by definition, $a_1, \dots, a_d$, so $c_0 \neq 0$ and is thus invertible. So one can define $k_0 := c_0^{-1}b_0$. Then

$$f(x) := k_0 g_0(x)$$

satisfies $f(a_0) = b_0, f(a_1) = f(a_2) = \dots = f(a_d) = 0$.

For the general case, proceed as follows: for all indices $i \in \{0, \dots, d\}$ such that $b_i \neq 0$, let

$$g_i(x) := \frac{1}{x - a_i} \prod_{j=0}^d (x - a_j),$$

i.e., $g_i(x) = (x - a_1) \cdots (x - a_{i-1})(x - a_{i+1}) \cdots (x - a_d)$. As in the previous paragraph, one can define

$$k_i := g_i(a_i)^{-1}b_i.$$

Now, if $i_1, \dots, i_t \in \{0, \dots, d\}$ are all the indices such that $b_{i_j} \neq 0$, then

$$f(x) := k_{i_1}g_{i_1}(x) + k_{i_2}g_{i_2}(x) + \dots + k_{i_t}g_{i_t}(x)$$

satisfies $f(a_0) = b_0, f(a_1) = b_1, \dots, f(a_d) = b_d$. Further, this is the unique polynomial to satisfy the desired property by Q7.